

AURORA: Autonomous-Uncertainty-aware-Resilient-Orbital-Reconfiguration-Architectur

Multi-Sensor Estimation, GNSS Fault Management, and AI-Assisted Anomaly Detection

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Abstract

A navigation estimate is useful only when its reliability can also be assessed. For a low Earth orbit (LEO) CubeSat, GNSS faults, reduced measurement redundancy, inertial-sensor errors, and temporary outages can compromise both state accuracy and confidence in that state. This paper presents AURORA, a modular Python simulation framework for resilient GNSS-based navigation under these conditions. The architecture combines orbital propagation, a simplified GPS-like constellation and pseudorange model, an orbital extended Kalman filter (EKF), attitude estimation with a multiplicative EKF (MEKF), body-frame inertial aiding, residual-based fault detection and isolation (FDI), supervisory reconfiguration, and an AI-assisted anomaly layer using Isolation Forest with temporal features. Evaluation spans Monte Carlo testing, a 64-scenario Latin-hypercube robustness campaign over 23 parameters, detector comparisons, sensitivity screening, a 108-case geometry sweep, and controlled outage experiments. In a representative campaign, protected navigation reduced mission position RMSE from 1.674 m to 0.664 m. The hybrid AI-assisted detector improved detection probability over baseline FDIR by 16.69 percentage points in a held-out campaign and 29.24 percentage points in a broader frozen-detector robustness campaign, with false-alarm probabilities near 2.5%. Controlled experiments also showed that navigation-EKF maturity before an outage and outage duration materially affect in-outage error. These findings support a layered design: classical FDIR and conservative reconfiguration remain responsible for protection, while AI contributes additional anomaly evidence. The study is simulation-based and does not claim flight qualification or certified integrity.

Keywords: CubeSat; LEO; GNSS; resilient navigation; integrity monitoring; EKF; MEKF; FDIR; sensor fusion; Isolation Forest; Monte Carlo; outage resilience.

1 Introduction

Autonomous spacecraft must estimate their state while operating with imperfect sensors and incomplete observations. GNSS can provide valuable navigation measurements in LEO, but measurements may become corrupted, unavailable, or insufficiently redundant for reliable fault diagnosis. In such cases, an estimator may still output a numerical position even though the available evidence no longer justifies treating that position as trustworthy.

This motivates the central distinction of AURORA: *navigation accuracy* describes how close an estimate is to truth, whereas *navigation integrity* concerns whether the system has adequate grounds to trust the estimate and recognize conditions in which it may be misleading. These concepts are related, but not interchangeable.

The research question is:

Can a LEO CubeSat maintain useful and trustworthy navigation during GNSS degradation, faults, and outages by combining multi-sensor state estimation with fault detection and supervisory reconfiguration, while using AI as complementary anomaly evidence?

1.1 Contributions

The project contributes: (1) a modular simulation chain connecting orbital dynamics, GNSS geometry and pseudoranges, inertial sensors, attitude estimation, and navigation estimation; (2) an orbital EKF and quaternion-based attitude MEKF; (3) residual-based GNSS FDI coupled to a conservative protected-navigation policy; (4) an AI-assisted anomaly supervisor using Isolation Forest and temporal features; (5) Monte Carlo, Latin-hypercube robustness, sensitivity, geometry, and outage campaigns; and (6) controlled studies separating outage duration from navigation-estimator maturity, with explicit treatment of confounding and model limitations.

2 System Architecture and Assumptions

2.1 Architecture

Figure 1 summarizes the functional layers. The estimator forms the state estimate; FDIR evaluates measurement consistency and fault observability; the resilience manager selects a conservative response. The AI supervisor supplies complementary anomaly evidence. AI non-detection is not interpreted as proof of integrity.

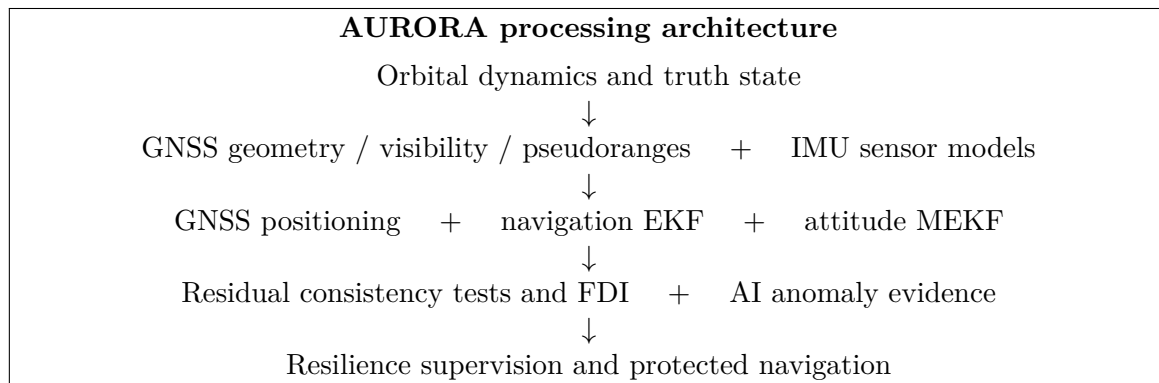


Figure 1: Functional architecture. AI augments anomaly evidence; it does not independently certify navigation integrity.

2.2 Reference scenario

The baseline orbit is approximately 550 km altitude. Propagation uses two-body gravity and, in selected studies, the J_2 perturbation. The GNSS environment is a simplified GPS-like constellation of 24 satellites in six planes. Visibility uses spherical-Earth line-of-sight occultation.

This is an algorithm-development environment, not a receiver-level digital twin. It omits realistic antenna patterns, carrier-to-noise ratio, acquisition/tracking limits, detailed spacecraft body blockage, channel limits, real broadcast ephemerides, satellite clock errors, ionospheric delay, and full Earth-orientation and signal-propagation effects. Numerical results are conditional on this modeled environment.

3 Models and Estimation

3.1 Orbital propagation

The Cartesian state is

$$\mathbf{x} = \begin{bmatrix} \mathbf{r}^T & \mathbf{v}^T \end{bmatrix}^T, \quad (1)$$

with two-body dynamics

$$\dot{\mathbf{r}} = \mathbf{v}, \quad \dot{\mathbf{v}} = -\frac{\mu}{\|\mathbf{r}\|^3} \mathbf{r}. \quad (2)$$

Selected experiments augment acceleration with a J_2 term. The force model is not intended as a complete high-fidelity orbit model.

3.2 GNSS pseudorange and geometry

A simplified pseudorange is

$$\rho_i = \|\mathbf{r}_i - \mathbf{r}\| + cb + \epsilon_i, \quad (3)$$

where $\mathbf{r}$ and $\mathbf{r}_i$ are receiver and satellite positions, b is receiver clock bias, and ϵ_i includes modeled noise and injected faults. The unweighted geometry matrix has rows $H_i = [-u_x, -u_y, -u_z, 1]$, under the project's sign convention. With $Q = (H^T H)^{-1}$,

$$PDOP = \sqrt{Q_{xx} + Q_{yy} + Q_{zz}}. \quad (4)$$

PDOP summarizes position geometry under this simplified unweighted model; it does not itself measure fault detectability or guarantee integrity.

3.3 Navigation EKF

The EKF prediction is

$$\hat{\mathbf{x}}_k^- = f(\hat{\mathbf{x}}_{k-1}), \quad P_k^- = F_k P_{k-1} F_k^T + Q_k. \quad (5)$$

For measurement $\mathbf{z}_k$, innovation and covariance are

$$\mathbf{y}_k = \mathbf{z}_k - h(\hat{\mathbf{x}}_k^-), \quad S_k = H_k P_k^- H_k^T + R_k. \quad (6)$$

The gain and state update are

$$K_k = P_k^- H_k^T S_k^{-1}, \quad \hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + K_k \mathbf{y}_k. \quad (7)$$

The normalized innovation squared (NIS),

$$NIS = \mathbf{y}_k^T S_k^{-1} \mathbf{y}_k, \quad (8)$$

is used as a consistency diagnostic. Position error alone is not sufficient to demonstrate statistical consistency.

3.4 Inertial aiding and attitude

The accelerometer and gyro models are

$$\mathbf{f}_{m,B} = \mathbf{f}_{true,B} + \mathbf{b}_{a,B} + \mathbf{n}_{a,B}, \quad (9)$$

$$\boldsymbol{\omega}_m = \boldsymbol{\omega}_{true} + \mathbf{b}_g + \mathbf{n}_g. \quad (10)$$

The attitude MEKF propagates quaternion attitude with gyro measurements and corrects it using synthetic star-tracker observations; quaternion-based spacecraft attitude estimation is a standard formulation [?]. Body-to-inertial rotation transforms body-frame specific force for inertial navigation. Bias states are included in selected variants. The coupled navigation covariance does not represent a fully joint attitude-navigation covariance, limiting uncertainty-consistency claims.

4 Fault Detection, Isolation, and Reconfiguration

4.1 Residual detection

Pseudorange residuals are

$$r_i = \rho_{\text{measured},i} - \rho_{\text{predicted},i}. \quad (11)$$

Under the project’s simplified equal-noise assumption, a global statistic is

$$T = \frac{\mathbf{r}^T \mathbf{r}}{\sigma_\rho^2}. \quad (12)$$

The statistic is compared with a threshold to identify inconsistency, following the residual-monitoring principle used in classical GPS integrity monitoring [?]. Its interpretation depends on the noise model, degrees of freedom, and geometry; unequal weighting requires an appropriate weighted residual covariance.

4.2 Leave-one-out isolation

After a global alarm, leave-one-out tests remove one satellite observation at a time and recompute consistency. A candidate is isolated when its removal restores consistency according to the configured test. A four-unknown position-and-clock solution with four measurements has no residual degrees of freedom. AURORA therefore uses $N \geq 6$ as a conservative project-specific protection gate, not a universal GNSS rule.

Fault observability depends on how a fault projects into the residual subspace. More visible satellites or low PDOP do not guarantee equal fault sensitivity in every geometry.

4.3 Resilience manager

The resilience manager mediates between navigation availability and trust. The permissive configuration accepts a broader set of conditions; protected operation applies stricter redundancy and fault-management logic. The objective is not to force a nominal-looking answer at all costs, but to avoid treating insufficiently supported measurements as unquestionably trustworthy.

5 AI-Assisted Anomaly Detection

The AI layer uses Isolation Forest, an unsupervised anomaly-detection method based on random partitioning [?], with features derived from GNSS residual and temporal behavior. Temporal features allow the detector to respond to patterns across samples rather than only instantaneous residuals. A hybrid supervisor combines AI anomaly evidence with classical detection evidence.

The division of responsibility is deliberate. Classical residual-based FDIR and the resilience policy remain the primary protection mechanisms. AI can improve detection sensitivity in the tested datasets, but an AI non-alarm does not clear a fault and is not an integrity certificate. Detection probability must be interpreted alongside false-alarm probability.

6 Experimental Design

6.1 Monte Carlo and robustness

Monte Carlo trials repeat controlled scenarios with different stochastic realizations, allowing performance distributions to be characterized instead of relying on one run. The campaigns include nominal operation, injected GNSS faults, low-redundancy stress cases, and sensor/GNSS outages.

A Latin hypercube campaign evaluated 64 scenarios across 23 parameters. Sensitivity analysis was a screening analysis based on correlation/rank relationships; it was not a global Sobol decomposition and does not establish causality.

6.2 Geometry and outage studies

The geometry campaign covered 108 combinations of three altitudes, six inclinations, and six orbital phases. Further campaigns varied outage phase, outage duration, and navigation-EKF maturity before outage onset. A combined campaign varied maturity and duration together.

The delayed-EKF experiment activates the navigation filter later and initializes it from truth plus a fixed prior error. It is a controlled truth-referenced simulation intervention, not a flight-realistic onboard restart design. Repeated-measures non-parametric tests were used for matched experiments, with multiple-comparison corrections applied where specified.

7 Results

7.1 Protected navigation

Table 1 reports a representative Phase 16 comparison. Protected navigation reduced mission RMSE by approximately 60% relative to the permissive configuration.

Table 1: Representative protected-navigation campaign; results are conditional on the specified simulated scenario.

Metric	Permissive	Protected
Mission position RMSE [m]	1.674	0.664
Low-redundancy RMSE [m]	4.054	0.416
Dual-outage RMSE [m]	1.665	0.408
Mean NIS	8.391	3.013

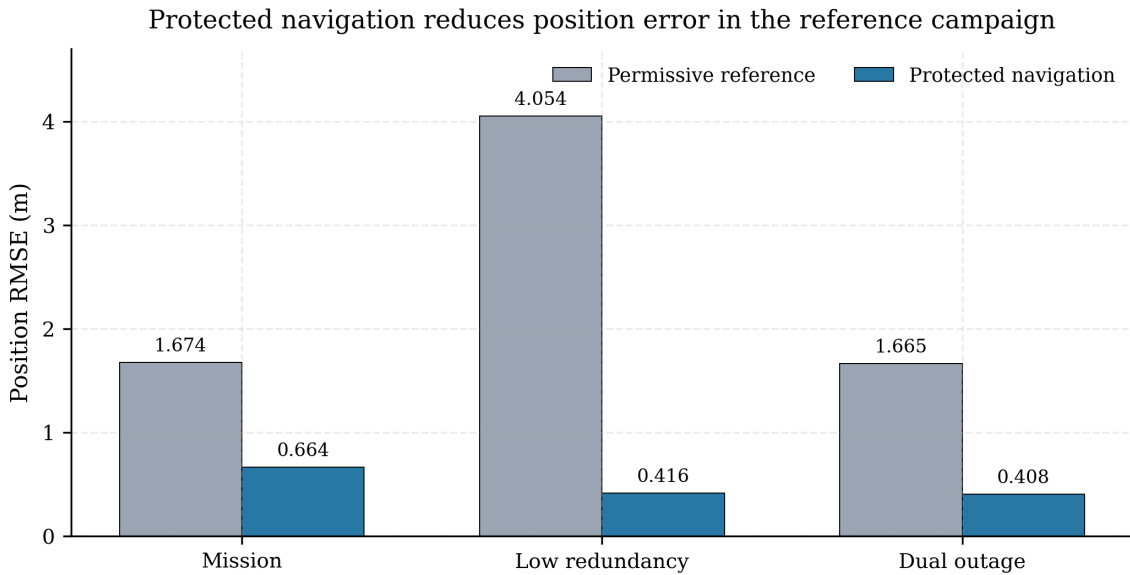


Figure 2: Position RMSE for permissive and protected configurations across the three reported conditions. Values are aggregate results from the Phase 16 reference robustness campaign.

Strong-fault detection/isolation was approximately 99.97%, low-redundancy detection was 34.34%,

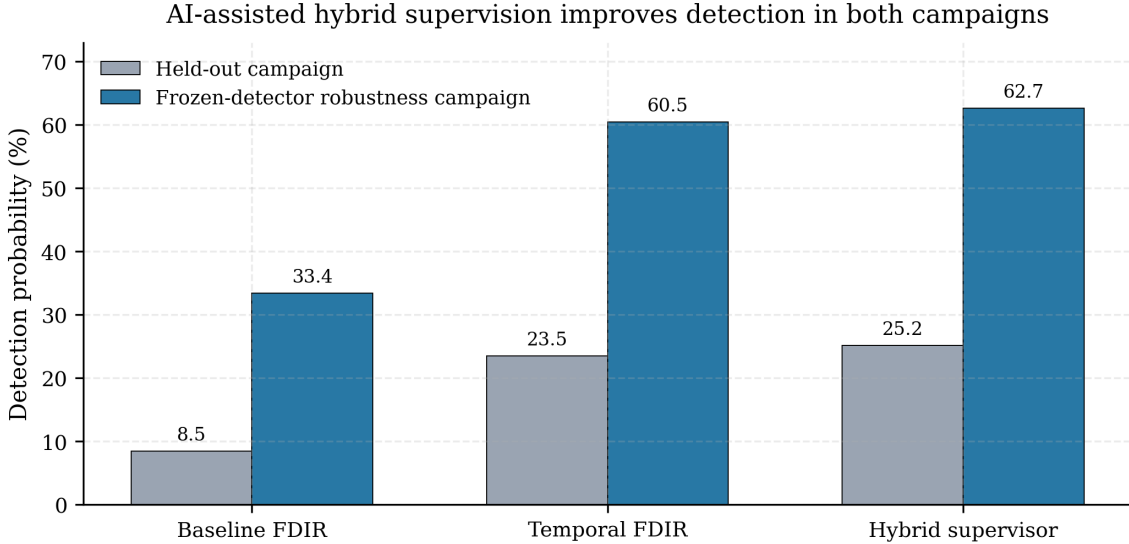
and false alarms were 0.99% in this campaign. The contrast shows why detector performance must be tied to fault magnitude, redundancy, geometry, and false-alarm rate rather than summarized by a single headline percentage.

7.2 AI-assisted detection

Table 2 reports two separate detector campaigns. The results should not be pooled because the test conditions differ.

Table 2: AI-assisted detection results in two separate campaigns.

Metric	Held-out campaign	Robustness campaign
Baseline FDIR detection [%]	8.48	33.41
Temporal FDIR detection [%]	23.52	60.45
Hybrid detection [%]	25.17	62.66
Hybrid gain over baseline [pp]	16.69	29.24
Hybrid false alarms [%]	2.45	2.47



Hybrid false-alarm probability: 2.45% (held-out) and 2.47% (robustness). Campaigns are distinct and not pooled.

Figure 3: Detection probability for baseline FDIR, temporal FDIR, and the hybrid supervisor in two separate evaluation campaigns. The campaigns have different test conditions and are shown side by side, not pooled.

The hybrid supervisor improved detection probability in both evaluations, but the results do not establish generalization to real flight telemetry. The AI layer is best interpreted as complementary evidence.

7.3 Robustness and sensitivity

Pseudorange noise was a dominant correlate of protected-navigation RMSE: $\rho \approx 0.842$ for mission RMSE, $\rho \approx 0.520$ for protected low-redundancy RMSE, and $\rho \approx 0.428$ for dual-outage RMSE. Fault-to-noise ratio correlated strongly with detection performance ($\rho \approx 0.823$ for FDIR and $\rho \approx 0.860$ for the hybrid detector). These are screening associations within the sampled domain, not causal effects or universal parameter rankings.

7.4 GNSS geometry and fault observability

Across 108 simplified geometry cases, mean visible-satellite count was 13.79 and mean PDOP was 1.031; minimum natural visibility was 12 satellites. Five-satellite cases were therefore deliberate low-redundancy stress tests, not natural outputs of this simplified constellation model.

Protected navigation was comparatively stable across representative geometries, while fault detection in five-satellite conditions varied substantially. This supports a key distinction: PDOP describes position sensitivity in the modeled geometry, whereas residual-based fault detection depends on fault projection into residual space and the available degrees of freedom.

7.5 Outage phase

An eight-phase sweep with eight replicates per phase tested a fixed 60–75 second outage. The Friedman test found no significant orbital-phase effect on outage RMSE ($p \approx 0.731$) or terminal error ($p \approx 0.635$). Pre-outage PDOP varied with phase, but its correlation with outage RMSE was $\rho \approx 0.015$ ($p \approx 0.904$). In this setup, instantaneous phase/PDOP was not the dominant predictor of autonomous drift.

7.6 Estimator maturity before outage

The observational outage-timing study suggested that later outages tended to have lower start uncertainty and lower outage error, but timing and estimator maturity were confounded. Partial correlations controlling for mission time were not statistically significant, so this observational study alone does not establish a causal mechanism.

A controlled maturity experiment held outage duration fixed while varying navigation-EKF age. Mean outage RMSE decreased from 4.476 m at 5 minutes of maturity to 0.802 m at 60 minutes. The global Friedman test was significant ($p \approx 3.31 \times 10^{-4}$). Pairwise comparisons did not all remain significant after Holm correction; the supported claim is a global repeated-measures effect across the tested conditions, not that every pair differs significantly.

Table 3: Confirmed endpoint values from the controlled EKF-maturity experiment.

Estimator age before outage	Outage RMSE [m]	Terminal error [m]
5 min	4.476	7.570
60 min	0.802	1.179

The intervention initializes the activated filter from truth plus a fixed prior error, so it isolates prior assimilation in simulation but does not represent an onboard restart architecture. Intermediate values are omitted here because the consolidated project summary confirms the endpoints and global test, not every intermediate row.

7.7 Outage duration and interaction with maturity

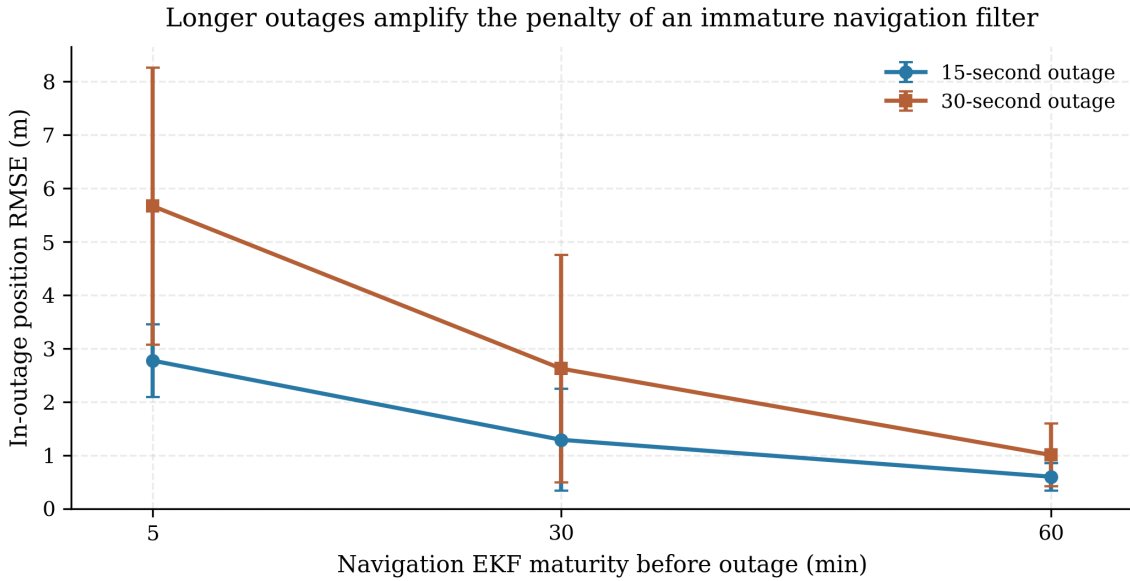
With a mature estimator, longer outages increased in-outage RMSE.

Table 4: In-outage RMSE as a function of outage duration for a mature estimator.

Outage duration [s]	Mean outage RMSE [m]
5	0.396
10	0.485
15	0.611
20	0.781
30	1.257

The repeated-measures duration effect was significant ($p \approx 2 \times 10^{-6}$). Post-recovery RMSE over the tested recovery window did not show a significant duration effect ($p \approx 0.121$). The evidence supports increasing error during the outage, but not a claim that recovery error necessarily worsens monotonically.

In the combined maturity-duration campaign, a 15-second outage produced RMSE values of 2.773 m, 1.291 m, and 0.599 m for estimator ages of 5, 30, and 60 minutes. For a 30-second outage, the values were 5.667 m, 2.624 m, and 1.005 m. The outage-RMSE interaction diagnostic remained significant after the specified Holm correction across interaction metrics ($p \approx 0.022$). This supports an interaction within the tested setup, not a claim that every age-specific pairwise contrast is independently significant.



Markers show mean; error bars show ± 1 standard deviation across 8 runs per condition.

Figure 4: Combined Phase 17H experiment: mean in-outage position RMSE versus navigation-EKF maturity for 15-second and 30-second outages. Error bars denote one standard deviation across eight runs per condition.

Figure 4 makes the interaction visible: the longer outage is associated with higher error at each tested maturity, and the gap is largest for the least mature estimator. These are finite simulated conditions, not flight-performance limits.

8 Discussion

Protection must be distinct from estimation. An EKF can propagate a state during degraded observations, but the state estimate alone does not establish whether the measurements are trust-

worthy. Separating estimation from FDIR and supervisory reconfiguration makes this distinction explicit.

Redundancy affects diagnosability. The strong-fault and low-redundancy results show that detector performance depends on geometry and available degrees of freedom. Detection rates must be reported with scenario and false-alarm context.

AI can improve sensitivity without becoming the integrity authority. The hybrid supervisor improved detection probability in both evaluated campaigns. Its false alarms and dependence on the simulation domain mean it should complement, not replace, classical integrity logic.

Resilience depends on outage onset conditions. Longer prior assimilation improved the filter’s starting condition in controlled experiments, while longer outages allowed error to accumulate. Outage duration alone is therefore insufficient to characterize resilience.

Accuracy is not certification. The sub-metre RMSE is a result under specified simulated conditions, not a guarantee for real operation and not a certified integrity-risk bound.

9 Limitations and Validity

The work is simulation-based and bounded by the following limitations:

1. The constellation and visibility model are simplified; tested visibility is not representative of every receiver or mission.
2. Antenna patterns, CN_0 , acquisition/tracking, channel limits, detailed body blockage, and real ephemerides are not modeled.
3. Satellite clock, ionospheric, relativistic, Sagnac, and Earth-orientation effects are not implemented at operational fidelity.
4. Force and sensor models are simplified or synthetic; the coupled covariance omits full attitude-navigation cross-covariance.
5. FDIR thresholds depend on specified noise assumptions; unequal weighting and non-Gaussian errors require recalibration.
6. AI was evaluated on controlled simulation data, not independent flight telemetry.
7. Sensitivity results are screening correlations, not causal or global variance decompositions.
8. Geometry and outage results cover finite scenario sets; the resilience map is not a certified continuous performance envelope.
9. No hardware-in-the-loop, receiver-in-the-loop, or flight validation has been performed.

These limitations bound the claims but do not remove the value of controlled comparisons within the stated model.

10 Conclusion and Future Work

AURORA presents a layered simulation framework for resilient LEO CubeSat navigation under GNSS faults and outages. It combines orbital and GNSS models, EKF/MEKF estimation, inertial aiding, residual-based FDIR, supervisory reconfiguration, and AI-assisted anomaly evidence. In a representative campaign, protected navigation reduced mission RMSE from 1.674 m to 0.664 m. The hybrid supervisor improved detection probability over baseline FDIR by 16.69 and 29.24 percentage points in two separate campaigns. Robustness and sensitivity experiments identified pseudorange noise and fault-to-noise ratio as important performance correlates. Geometry and outage experiments showed that fault observability depends on redundancy and geometry, while estimator maturity and outage duration jointly influence in-outage performance.

Future work should cross-check the simplified orbital/GNSS environment against an independent simulator such as STK or GMAT or real ephemerides, add higher-fidelity receiver and spacecraft models, use real GNSS/IMU data, perform hardware-in-the-loop testing, and develop a more rigorous

integrity-risk framework. STK/GMAT cross-validation has not yet been performed and remains future work.

The central conclusion is that resilient navigation requires not only estimating the state, but also evaluating the evidence supporting that estimate and responding conservatively when the evidence is insufficient.

Reproducibility note

The project repository contains modular Python code, experiment scripts, tests, and saved result tables/figures. Before external submission, each reported number should be cross-checked against its committed CSV and experiment configuration. The final archived release should record software versions, seeds, thresholds, and scenario definitions.

Acknowledgement

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